CS498GC Mobile Robotics

Extra Credit Quiz Questions

Submission Guidelines

Earn up to +7 Extra Credit Points

+5 points per selected question
+2 bonus for TA-proof questions

Fall 2025

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Course Website: kulbir-singh-ahluwalia.com/cs498gc/fa25

Gradescope Code: KDP5G8

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Contents

1 Overview

This document outlines the requirements and guidelines for submitting extra credit quiz questions for CS498GC Mobile Robotics. Students have the opportunity to contribute challenging, AI-resistant questions for Quiz 2 and Quiz 3, earning significant extra credit points.

1.1 Extra Credit Structure

Component	Points
Base Extra Credit (per selected question)	+5 points
TA-Proof Bonus (exceptionally challenging)	+2 points
Maximum Points per Question	+7 points

Important

Only the **TOP 10 SUBMISSIONS** based on quality criteria will receive extra credit. Focus on creating one exceptional question rather than multiple mediocre ones.

2 Submission Deadlines

Deadline

Quiz 2 Questions

- Submission Deadline: **November 21, 2025 @ 11:00 PM**
- Quiz Date: November 21, 2025 (Online, 24-hour window)

Deadline

Quiz 3 Questions

- Submission Deadline: **December 9, 2025 @ 11:00 PM**
- Quiz Date: December 8, 2025 (Online, 24-hour window)

3 Submission Requirements

3.1 Question Format

All submissions must be **MULTIMODAL**, incorporating both visual and textual elements:

1. **Visual Components** (Required)
   - Diagrams, graphs, or charts
   - Mathematical equations as images
   - System architectures or flowcharts
   - Sensor data visualizations
   - Robot trajectory plots

2. **Text Components** (Required)
   - Clear problem statement
   - Context and background information
   - Multiple choice options (A, B, C, D, etc.)
   - Instructions for solving
3. **Answer Format** (Required)
   - Clearly specify: "Multiple Choice - Single Correct Answer" OR
   - "Multiple Choice - Multiple Correct Answers"
   - Include 4-6 answer options
   - Provide detailed solution explanation

3.2 Quality Criteria

Questions will be evaluated based on four primary criteria:

Criterion	Description	Weight
Difficulty	Requires deep understanding of course concepts	30%
Semantic Complexity	Multi-step reasoning and concept integration	25%
Creativity	Novel problem formulation or unique perspective	20%
Documentation	Professional presentation and clarity	25%

4 AI-Resistance Requirements

4.1 What Makes a Question AI-Proof?

Your question must demonstrate resistance to current AI tools including:

- GPT-4 and GPT-4 Vision
- Claude Opus 4.1
- Google Gemini
- Other multimodal LLMs

4.2 Effective Strategies

Success Tip

Strong AI-Resistant Elements:

1. **Visual-Spatial Reasoning:** Interpreting complex robot trajectories or sensor coverage maps
2. **Multi-Step Integration:** Problems requiring kinematics $\rightarrow$ dynamics $\rightarrow$ control
3. **Numerical Verification:** Where intermediate calculations affect the final answer
4. **Domain-Specific Context:** Real-world scenarios requiring robotics expertise
5. **Time-Series Analysis:** Interpreting sensor data over time from graphs

4.3 Example Topics

- Extended Kalman Filter with visual state evolution
- Path planning with complex obstacle configurations
- Sensor fusion requiring graphical data interpretation
- Control system stability from Bode/Nyquist plots
- SLAM visualization and loop closure detection
- Kinematic chain analysis with visual representations
- Multi-robot coordination scenarios

5 Submission Format on Gradescope

5.1 Required Files

Submit the following files to Gradescope under "Extra Credit Quiz Questions":

1. **question.pdf** - Main question document
    - Question text with embedded images/diagrams
    - Multiple choice options clearly labeled
    - Any necessary formulas or equations
    - Space for student work (if applicable)
2. **solution.pdf** - Complete solution
    - Correct answer(s) clearly marked
    - Step-by-step solution process
    - Explanation of why incorrect options are wrong
    - Justification for AI-resistance
3. **metadata.txt** - Question information
    - Question type (Single/Multiple correct)
    - Topic area (e.g., EKF, Path Planning)
    - Difficulty rating (1-10 scale)
    - AI-resistance explanation (200 words)

5.2 File Naming Convention

Use the following naming format:

`[YourNetID]_[Quiz2/Quiz3]_[Topic]_[question/solution/metadata].[pdf/txt]`

Example: `ksa5_Quiz2_EKF_question.pdf`

6 Evaluation Process

6.1 Selection Criteria

Questions will be evaluated in three stages:

1. **Initial Screening**
   - Format compliance check
   - Multimodal requirement verification
   - Basic quality assessment
2. **AI Testing**
   - Test with multiple AI models
   - Verify resistance claims
   - Check for solution uniqueness
3. **Final Selection**
   - Rank by quality criteria
   - Select top 10 submissions
   - Award extra credit points

6.2 TA-Proof Bonus

For the additional +2 points, questions must be:

- Challenging for graduate-level understanding
- Require extensive domain knowledge
- Have non-obvious solution paths
- Demonstrate exceptional creativity

7 Tips for Success

Success Tip

DO:

- Test your question with AI tools before submission
- Include realistic robotics scenarios
- Use high-quality, clear diagrams
- Provide unambiguous problem statements
- Consider edge cases in your solution
- Reference course material appropriately

> **Important**
>
> **DON'T:**
>
> - Submit text-only questions
> - Use easily searchable facts
> - Create ambiguous questions
> - Copy from existing resources
> - Submit questions with errors
> - Ignore formatting requirements

8 Example Question Structure

8.1 Sample Format

Question: Consider the mobile robot trajectory shown in Figure 1, where the robot uses an Extended Kalman Filter for localization...

[Complex diagram showing robot path, sensor readings, and uncertainty ellipses]

Given the following sensor measurements at time $t = 5s$:

- IMU: $\omega_z = 0.5$ rad/s
- GPS: $(x, y) = (10.2, 5.8)$ m
- Wheel encoders: $v_L = 1.2$ m/s, $v_R = 1.4$ m/s

What is the estimated position uncertainty after the measurement update?
Options:

A. $\sigma_x = 0.3m, \sigma_y = 0.5m$

B. $\sigma_x = 0.5m, \sigma_y = 0.3m$

C. $\sigma_x = 0.8m, \sigma_y = 0.6m$

D. $\sigma_x = 0.2m, \sigma_y = 0.2m$

9 Support and Resources

9.1 Getting Help

- **Campuswire:** Post with tag `#extra-credit-questions`
- **Office Hours:** Get feedback on question ideas
- **Email:** ksa5@illinois.edu for submission confirmation

9.2 Additional Resources

- Course website: https://kulbir-singh-ahluwalia.com/cs498gc/fa25/
- Gradescope Course ID: 1102297
- Entry Code: KDP5G8

10 Special Note: Quiz 1 Calculator Policy

For students affected by the Quiz 1 calculator policy:

We acknowledge that the "no digital devices" policy during Quiz 1 was not explicitly clear about scientific calculators. This extra credit opportunity is especially encouraged for students who were affected by this ambiguity.

Future Quiz Policy Clarification:

- ✔ Basic scientific calculators ARE allowed
- ✘ Prohibited: smartphones, tablets, smartwatches, laptops, smart glasses, internet-connected devices

Good luck with your submissions!

Let's create challenging, meaningful assessments together.